E0 230 Computational Methods of Optimization

Tutorial 1

2024

Reading Assignment

Familiarize yourself with the following linear algebra terms. There are several equivalent definitions or statements about the them. Read up on all of them.

1. Positive definite matrix.
2. Spectral theorem for symmetric matrices.
3. Singular value decomposition.

Questions

1. Is the product of two positive definite matrices positive definite? Why or why not?

> **Solution:** No. The product of two symmetric matrices are not necessarily symmetric (by counterexample). Find two PD matrices that does not multiply to give a symmetric matrix.

2. Suppose $x,\ y \in \mathbb{R}^n$. Prove the following statements. Note that $\|\cdot\|$ refers to the Euclidean norm if not otherwise specified.
   (a) $|\|x\| - \|y\|| \leq \|x - y\| \leq \|x\| + \|y\|$

> **Solution:**
> We start by writing $\|x-y\|^2 = x^Tx + y^Ty - 2x^Ty = \|x\|^2 + \|y\|^2 - 2x^Ty$. Next, by Cauchy-Schwartz (derived in class, get familiar with this inequality!) we have $-2\|x\|\|y\| \leq -2x^Ty \leq 2\|x\|\|y\|$. From this, it follows that
> $$\begin{aligned} &\|x\|^2 + \|y\|^2 - 2\|x\|\|y\| \leq \|x-y\|^2 \leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| \\ \Rightarrow &(|\|x\| - \|y\||)^2 \leq \|x-y\|^2 \leq (\|x\| + \|y\|)^2 \\ \Rightarrow &|\|x\| - \|y\|| \leq \|x-y\| \leq \|x\| + \|y\| \end{aligned}$$

   (b) $2x^Ty = \|x\|^2 + \|y\|^2 - \|x-y\|^2$

> **Solution:** Recall $\|x-y\|^2 = x^Tx + y^Ty - 2x^Ty = \|x\|^2 + \|y\|^2 - 2x^Ty$. Rearranging terms, we can prove the statement.

   (c) Suppose $B \in \mathbb{R}^{m\times n}$. Show that the function $d(B) = \sqrt{\mathrm{Tr}(B^TB)}$ is a norm (this is the Frobenius norm of a matrix, denoted by $\|\cdot\|_F$).

Solution: To show a function is a norm, we need to show three things: (1) $d(B) = 0 \Leftrightarrow B = 0$; (2) $d(sB) = |s|d(B)$; (3) $d(A + B) \leq d(A) + d(B)$. The second property holds by the linearity of the trace operator. To see that the first property holds, we observe that $B^T B$ is a symmetric positive semidefinite matrix; thus, if $\mathrm{Tr}(B^T B) = 0$ it follows that all the eigenvalues of $B^T B$ are zero, which only holds if B is a zero matrix. Alternatively, if we sum the diagonal elements of $B^T B$, we get $d(B) = \sqrt{\sum_i \sum_j |b_{ij}|^2}$, which can only be zero if all elements are zero. Last, for the triangle inequality, we see that

$$\begin{aligned} d(A+B)^2 &= \mathrm{Tr}((A+B)^T(A+B)) \\ &= \mathrm{Tr}(A^T A + B^T B + B^T A + A^T B) \\ &= \mathrm{Tr}(A^T A) + \mathrm{Tr}(B^T B) + \mathrm{Tr}(B^T A) + \mathrm{Tr}(A^T B) \\ &= \mathrm{Tr}(A^T A) + \mathrm{Tr}(B^T B) + 2\mathrm{Tr}(B^T A) \end{aligned}$$

Now, recall that

$$\mathrm{Tr}(A^T B) = \sum_i \sum_j a_{ij} b_{ij}.$$

Thus, we can emply the Cauchy-Schwarz inequality, giving us

$$\begin{aligned} d(A+B)^2 &= \mathrm{Tr}(A^T A) + \mathrm{Tr}(B^T B) + 2\mathrm{Tr}(B^T A) \\ &\leq \mathrm{Tr}(A^T A) + \mathrm{Tr}(B^T B) + 2\sqrt{\mathrm{Tr}(B^T B)\mathrm{Tr}(A^T A)} \\ &= d(B)^2 + d(A)^2 + 2d(A)d(B). \end{aligned}$$

Thus, it follows that $d(A + B) \leq d(A) + d(B)$.

3. Suppose $x \in \mathbb{R}$. Show that $e^x \geq 1 + x$.

Solution:

We start by taking the Taylor series: $e^x = \sum_i \frac{x^i}{i!}$. The Lagrange remainder for the linear approximation gives us $e^x = 1 + x + \frac{e^z}{2}x^2$ for some $z \in [0, x]$. Since $e^z x^2 \geq 0$, it follows that $e^x \geq 1 + x$.

We can also think of this as an optimization problem. Note that $e^x \geq 1 + x \Rightarrow e^x - 1 - x \geq 0 \Rightarrow \min_x e^x - 1 - x \geq 0$. So now, let $g(x) = e^x - 1 - x$. $g'(x) = e^x - 1$, which gives us $e^{x^*} = 1 \Rightarrow x^* = 0$. To check optimality, we employ the second order condtions. For that $g''(x) = e^x$ which is always positive. Thus, $g(x)$ is minimized at $x = 0$, and $g(0) = 0$. Thus, we prove the statement.

4. Suppose $f \in C^1_L$, where $L > 0$; that is, $\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2$. If $f \in C^1_L$, show that the functions $g(x) = \frac{L}{2}x^T x - f(x)$ and $h(x) = \frac{L}{2}x^T x + f(x)$ are convex. Then, show that $-\frac{L}{2}\|y - x\|^2 \leq f(y) - f(x) - \langle \nabla f(x), y - x\rangle \leq \frac{L}{2}\|y - x\|^2$.

Solution: If $f \in C^1_L$, then we have

$$\begin{aligned} &\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2 \\ \Rightarrow\ &\|\nabla f(x) - \nabla f(y)\|_2 \|x - y\|_2 \leq L\|x - y\|_2^2 \\ \Rightarrow\ &|\langle \nabla f(x) - \nabla f(y), x - y\rangle| \leq L\langle x - y, x - y\rangle \quad \text{(by the Cauchy-Schwarz Inequality)} \\ \Rightarrow\ &\langle (Lx - \nabla f(x)) - (Ly - \nabla f(y)), x - y\rangle \geq 0 \ \text{ and } \ \langle (Lx + \nabla f(x)) - (Ly + \nabla f(y)), x - y\rangle \geq 0. \end{aligned}$$

By inspection, we see that $\nabla g(x) = L/2x^T x - \nabla f(x)$ and $\nabla f(x) = L/2x^T x + \nabla f(x)$, and are monotonic; thus, $g(x)$ and $h(x)$ are convex. To prove the final two inequalities, substitute $g(x)$ and $h(x)$ into the inequality

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

and rearrange terms (for full marks, this should be shown explicitly).

5. Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is twice differentiable, such that $0 < m \leq \lambda_i(H_f(x))$ for all i, x, where $\lambda_i(H_f(x))$ is the ith largest eigenvalue of $H_f(x)$. Show that there exists a constant l such that, for each x, y, we have

$$\frac{l}{2}\|y - x\|^2 \ + f(y) - f(x) - \nabla f(x)^T (y - x) \geq 0$$

Solution: We apply the Taylor series expansion to $f(y)$ centered at x:

$$\begin{aligned}
f(y) &= f(x) + \nabla f(x)^T (y - x) + \frac{1}{2}(y - x)^T H_f(x)(y - x) + o(\|x - y\|^2) \\
&= f(x) + \nabla f(x)^T (y - x) + \frac{1}{2}(y - x)^T H_f(z)(y - x) \text{ for some } z \\
&\geq f(x) + \nabla f(x)^T (y - x) + \frac{1}{2}(y - x)^T \left(\lambda_{\min}(H_f(z))I\right)(y - x) \\
&\geq f(x) + \nabla f(x)^T (y - x) + \frac{1}{2} \min_{i,w} |\lambda_i(H_f(w))| \|y - x\|^2 \\
&= f(x) + \nabla f(x)^T (y - x) + \frac{m}{2}\|y - x\|^2
\end{aligned}$$

6. Show that the function $h(x, y) = x^4 - 4xy + y^4$ has a global minimum. What is that global minimum?

Solution: $h(x, y)$ can be written as,

$$h(x, y) = x^4 + y^4 \left(1 - \frac{4xy}{x^4 + y^4}\right)$$

As $\|(x, y)\| = \sqrt{x^2 + y^2} \to \infty$, the term $\frac{4xy}{x^4+y^4} \to 0$. Hence,

$$\lim_{\|(x,y)\| \to \infty} h(x, y) = \lim_{\|(x,y)\| \to \infty} (x^4 + y^4) = +\infty$$

. Thus, h is coercive, and therefore has a global minimum.

To find the global minimum, we find the stationary points (by finding points where the gradient is 0). We find these points to be $(0, 0)$, $(1, 1)$, $(-1, -1)$. Of these points, we see that $h(1, 1) = -2$, which is the global minimum. This can be verified by checking the Hessian at that point, as well as the AM-GM inequality.

7. Suppose we have a function $f : \mathbb{R}^n \to \mathbb{R}$, that satisfies $\max_x |\lambda_{\max}(H_f(x))| \leq L$. Suppose $A \in \mathbb{R}^{n \times n}$. Can you provide an exact expression for $f(Ax) - f(x)$? Can you provide an upper bound for $f(Ax) - f(x)$?

Solution: Taylor series expansions:

$$f(Ax) = f(x) + \nabla f(x)^T((A-I)x) + o(\|(A-I)x\|) \tag{1}$$

$$f(Ax) = f(x) + \nabla f(x)^T((A-I)x) + \frac{1}{2}x^T Q(z)x \tag{2}$$

$$\tag{3}$$

where $Q(z) = (A-I)^T H(z)(A-I)$ for some z on the line segment between x and Ax. We then get

$$f(Ax) - f(x) = \nabla f(x)^T((A-I)x) + \frac{1}{2}x^T Q(z)x \tag{4}$$

$$\leq \nabla f(x)^T((A-I)x) + \frac{1}{2}(\max_i |\lambda_i(H(z))|)(H(z))x^T(A-I)^T(A-I)x \tag{5}$$

$$\leq \nabla f(x)^T((A-I)x) + \frac{L}{2}x^T(A-I)^T(A-I)x \tag{6}$$

8. Suppose we have positive definite $A \in \mathbb{R}^{n\times n}$, and linearly independent vectors $\{v_i\}_{i=1}^m$, where $m < n$. How would you convert the problem

$$\arg\min_x x^T Ax \text{ such that } x \in \text{span}\,(v_1, ..., v_m)$$

into an unconstrained problem? Does this problem have a unique solution? If so, under what conditions would this problem not have a unique solution?

Solution:
We have $f(x) = x^T Ax$. If we write $x = \sum_i \alpha_i v_i = V\alpha$, where the ith column of V is v_i, we get

$$f(x) = g(\alpha) = (V\alpha)^T A\,(V\alpha) = \alpha^T V^T AV\alpha.$$

Next, since $A \succ 0$, it follows that $V^T AV \succ 0$ as well, since $f(x) > 0 \Leftrightarrow g(\alpha) > 0$. Thus, $g(\alpha)$ has a unique minimum as well. The problem would not have a unique solution if A is PSD and if at least 1 v_i is in the nullspace of A.

9. Suppose we have matrices $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{m\times n}$, $m \neq n$

   (a) Suppose A has n orthogonal eigenvectors. Show that we can write $A = V^T\Lambda V$, where the columns of V are the eigenvectors of A, and the diagonal elements of Λ are the eigenvalues of A. When are we unable to use this decomposition? Verify this with the matrices

$$A_1 = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \quad A_2 = \begin{bmatrix} 2 & 0.5 \\ 1 & 3 \end{bmatrix}$$

Solution: Let (λ_i, v_i) be an eigenpair of A. Then, we have $Av_i = \lambda_i v_i$. Taking the matrix $V = [v_1, ..., v_n]$, it follows that $AV = V\Lambda$ where Λ is a diagonal matrix with $\Lambda_{ii} = \lambda_i$. Next, recall that $V^T = V^{-1}$ since the columns are orthogonal. Thus, it follows that $AVV^T = V\Lambda V^T = A$.

For the numerical examples, we see that A_1 is symmetric, and thus has orthogonal eigenvectors. Thus, we can find such a decomposition, where

$$A_1 = V_1 D_1 V_1^T; \quad V_1 = \begin{bmatrix} 1.382 & 0 \\ 0 & 3.618 \end{bmatrix}.$$

A_2 on the other hand is not symmetric, and does not have orthogonal eigenvectors. We get

$$A_2 = V_2 D_2 V_2^T; \quad V_2 = \begin{bmatrix} -.807 & -.344 \\ .591 & -.939 \end{bmatrix}.$$

Note that V_2 still has full rank (in this case, 2), and we can still diagonalize A_2 with the transformation $A_2 = V_2 D_2 V_2^{-1}$ (why?).

(b) Suppose B is of rank r. What are the ranks of BB^T and $B^T B$? With this result, what can you say about the row and column ranks of B and B^T?

Solution:
From the rank-nullity theorem, we have $\text{rank}(B) + \text{nullity}(B) = \min\{m, n\}$. Thus, $Bx = 0 \Rightarrow B^T Bx = 0$, since $B^T Bx = 0 \Rightarrow x^T B^T Bx = (Bx)^T Bx = 0$. Thus, $\text{rank}(B) = \text{rank}(B^T B)$. Similarly, we can show that $B^T y = 0 \Rightarrow BB^T y = 0 \Rightarrow \text{rank}(B^T) = \text{rank}(BB^T)$. Next, we know that the columns of $B^T B$ are linear combinations of the columns of $B^T \Rightarrow \text{rank}(B^T B) \leq \text{rank}(B^T)$. Similarly, the columns of BB^T are linear combinations of the columns of $B \Rightarrow \text{rank}(BB^T) \leq \text{rank}(B)$. Thus, we get $\text{rank}(B) = \text{rank}(B^T B) \leq \text{rank}(B^T)$ and $\text{rank}(B^T) = \text{rank}(BB^T) \leq \text{rank}(B)$. This only holds if $\text{rank}(B) = \text{rank}(B^T)$.

(c) Show that we can decompose $B = U\Sigma V^T$, where Σ is diagonal and positive-semidefinite, and U and V have orthogonal columns. The diagonal elements of Σ are known as the singular values - how would you define them? Verify these results with the matrix

$$B = \begin{bmatrix} 4 & -1 & 5 \\ 1 & 3 & -2 \end{bmatrix}.$$

Solution: Let V be a matrix containing the eigenvectors of nonzero eigenvalues of $B^T B$ and let U be a matrix containing the eigenvectors of nonzero eigenvalues of BB^T. Let the eigenvalues be σ_i^2. Furthermore, we can write $u_i = \frac{1}{\sigma_i} B v_i$ (why? $BB^T u_i = BB^T(Bv_i)/\sigma_i = \sigma_i^2 B v_i / \sigma_i$). From this, it folows we can write $U = BV\Sigma^{-1} \Rightarrow U\Sigma = BV \Rightarrow B = U\Sigma V^T$ since $VV^T = I$.

To solve the numerical problem, we can use standard tools in python (`numpy.linalg.svd(·)`) or MATLAB (`svd(·)`).

(d) Let $B \in \mathbb{R}^{m \times n}$ be of rank $n < m$. Show that $\|B\|_F = \sqrt{\sum_k \sigma_k(B)^2}$, where $\sigma_k(B)$ is the kth largest singular value of B. Using this result, show that $\|B\|_F^2 \geq n \left(\det(B^T B)\right)^{\frac{1}{n}}$. Verify these results with $\hat{B} = B^T$ from the previous example (in this example, $n = 2$ and $m = 3$).

Solution: We can write $B = U\Sigma V^T$. Then, we have

$$\begin{aligned} \|B\|_F &= \sqrt{\text{Tr}\,(B^T B)} = \sqrt{\text{Tr}\,(V\Sigma^T U^T U \Sigma V^T)} \\ &= \sqrt{\text{Tr}\,(V\Sigma^T \Sigma V^T)} = \sqrt{\text{Tr}\,(\Sigma^T \Sigma V^T V)} = \sqrt{\text{Tr}\,(\Sigma^2)} \\ &= \sqrt{\sum_k \sigma_k^2}. \end{aligned}$$

More simply, the singular values of B are the eigenvalues of $B^T B$; thus, by the definition of trace, the expression holds.

For the second statement to be proved, we use the AM-GM inequality as follows:

$$\|B\|_F^2 = \sum_k \sigma_k(B) \geq n \left(\prod_k \sigma_k(B) \right)^{\frac{1}{n}} = n \det(BB^T)^{\frac{1}{n}}.$$

For the numerical example, we see that the Frobenius norm of $\|B\|_F^2 = 56$, whereas $2\sqrt{\det(BB^T)} = 26\sqrt{3} \leq 56$.

(e) Consider the function $f : U \to \mathbb{R}$, where $f(x) = x^T Ax$ and U is the set of all unit vectors of dimension n. Show that, if A is symmetric, range$(f) \subseteq [\lambda_{min}(A), \lambda_{max}(A)]$.

Solution:
Note that since A is symmetric, we can write $A = Q\Lambda Q^T$, where Q is an orthogonal matrix. Furthermore, let $y = Qx$, and note that $\|y\| = \sqrt{x^T Q^T Qx} = \sqrt{x^T x} = \|x\| = 1$. Thus, $f(x) = x^T Q^T \Lambda Qx = y^T \Lambda y = \lambda_1 y_1^2 + ... + \lambda_n y_n^2$. Then, it follows that $\lambda_{\min} \leq \lambda_1 y_1^2 + ... + \lambda_n y_n^2 \leq \lambda_{\max}$

(f) What is the solution to

$$s^* = \max_{x:\|x\|_2=1} x^T Bx,$$

where B is symmetric? *R*emark: s^* is the Spectral norm of B, denoted by $\|B\|_2$.

Solution:
Note that since B is symmetric, we can write $B = Q\Lambda Q^T$, where Q is an orthogonal matrix. Furthermore, let $y = Qx$, and note that $\|y\| = \sqrt{x^T Q^T Qx} = \sqrt{x^T x} = \|x\| = 1$. Thus, $x^T Q^T \Lambda Qx = y^T \Lambda y = \lambda_1 y_1^2 + ... + \lambda_n y_n^2 \leq (y_1^2 + ... + y_n^2)\lambda_{\max} = \lambda_{\max}$.

10. Suppose A, B are symmetric and that the problems

$$\text{(P1)} \quad \arg\min_x x^T Ax \quad \text{and} \quad \text{(P2)} \quad \arg\min_x x^T Bx$$

have unique solutions $x_{P1} = x_{P2} = 0$. What is the solution to

$$\arg\min_x x^T ABx,$$

and is it unique? (hint: every symmetric PD matrix has a unique, positive definite square root - can you prove this?)

Solution:
We know that P1 and P2 have unique solutions; thus, from the first and second order conditions of optimality, it follows that A, B are positive definite. We know that symmetric PD matrices have a PD square root, since each symmetric PD matrix C can be decomposed as $C = V\Lambda V^T = V\Lambda^{1/2}V^T V\Lambda^{1/2}V^T$. Thus, $C^{1/2} = V\Lambda^{1/2}V^T$. Then, we know that the spectrum of B is the same as the spectrum of PAP^{-1} for any invertible matrix P. Thus, the eigenvalues of B are the same as the eigenvalues of $C = A^{1/2}BA^{-1/2} \succ 0$. Thus, we see that $AB = A^{1/2}CA^{1/2}$, and $x^T ABx = y^T Cy$ where $y = A^{1/2}x$. Since C is PD, $\arg\min_y y^T Cy = x_{P2} = 0$.

11. Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is a convex function; that is, for each x, y and each $\alpha \in [0, 1]$, we have

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y).$$

Prove the following.

(a) Show that if f is convex, then every local minimum is a global minimum as well.

Solution: Let x^* be a local minimum, and suppose $f(x*) \leq f(x)\ \forall\ x \in B_\epsilon(x^*)$ for some $\epsilon > 0$. Next, let y be a point such that $f(y) < f(x^*)$. By the convexity of f, we have

$$\begin{aligned} f(\alpha x^* + (1-\alpha)y) &\leq \alpha f(x^*) + (1-\alpha)f(y) \\ &\leq f(x^*) \end{aligned}$$

for all $\alpha in(0,1]$. However, as $\alpha \to 1$, we have $z_\alpha = \alpha x^* + (1-\alpha)y \to x^*$ and $f(z_\alpha) < f(x^*)$, which violates the local optimality of x^*. Thus, we prove the statement by that contradiction.

(b) Show that if f is convex, then for each x, y, we have

$$f(y) \geq f(x) + \nabla f(x)^T(y-x).$$

Solution: First, we prove that this holds for the 1-dimensional case , and then generalize this result to the higher dimensional setting.
First, we prove $1 \Rightarrow 2$. From the definition of convexity, we have

$$\begin{aligned} f((1-\lambda)x + \lambda y) &\leq \lambda f(y) + (1-\lambda)f(x) \\ &= f(x) + \lambda\left(f(y) - f(x)\right). \end{aligned} \tag{7}$$

We rearrange terms, giving us

$$\begin{aligned} f((1-\lambda)x + \lambda y) - f(x) &\leq \lambda\left(f(y) - f(x)\right) \\ \frac{f((1-\lambda)x + \lambda y) - f(x)}{\lambda} &\leq f(y) - f(x) \\ \frac{f((1-\lambda)x + \lambda y) - f(x)}{\lambda} + f(x) &\leq f(y) \\ \frac{f((1-\lambda)x + \lambda y) - f(x)}{\lambda(y-x)}(y-x) + f(x) &\leq f(y) \end{aligned} \tag{8}$$

We then take the limit as λ approaches 0 from the right of (8).

$$\begin{aligned} &\lim_{\lambda \to 0^+} \frac{f((1-\lambda)x + \lambda y) - f(x)}{\lambda(y-x)}(y-x) + f(x) \leq f(y) \\ &\Rightarrow f'(x)(y-x) + f(x) \leq f(y). \end{aligned} \tag{9}$$

Hence, we have shown that $1 \Rightarrow 2$.
To show that $2 \to 1$, fix $x, y \in \mathbb{R}$ and $\lambda \in (0,1)$. Suppose $z = (1-\lambda)x + \lambda y$. Then, we have

$$\begin{aligned} f(x) &\geq f(z) + f'(z)(x-z) \\ f(y) &\geq f(z) + f'(z)(y-z) \\ \Rightarrow (1-\lambda)f(x) &\geq (1-\lambda)\left(f(z) + f'(z)(x-z)\right)) \\ \lambda f(y) &\geq \lambda\left(f(z) + f'(z)(y-z)\right). \end{aligned}$$

We then add the latter two expressions to get

$$\begin{aligned} (1-\lambda)f(x) + \lambda f(y) &\geq f(z) + f'(z)\left(\lambda(y-z) + (1-\lambda)(x-z)\right) \\ &\geq f(z) + f'(z)\left(\lambda(1-\lambda)(y-x) + (1-\lambda)\lambda(x-y)\right) \\ (1-\lambda)f(x) + \lambda f(y) &\geq f(z). \end{aligned}$$

Hence, we have shown that $2 \Rightarrow 1$ for the 1 dimensional case.
To prove the case for $f \in C^1(\mathbb{R}^n, \mathbb{R})$, define

$$g(t) = f(x + t(y - x)). \tag{10}$$

We can think of g as the restriction of f to the line segment connecting x and y. Then, we have

$$g'(t) = \frac{d}{dt} g(t) = \langle \nabla f(x + t(y - x)), y - x \rangle$$

. Since f is convex, we can show that g is convex. To do so, let $\alpha \in [0, 1]$ and write

$$\begin{aligned} g(\alpha t_1 + (1 - \alpha) t_2) &= f(x + (\alpha t_1 + (1 - \alpha) t_2)(y - x)) \\ &= f(x + \alpha t_1 (y - x) + (1 - \alpha) t_2 (y - x)) \\ &= f(\alpha(x + t_1(y - x)) + (1 - \alpha)(x + t_2(y - x))) \\ &\leq \alpha f(x + t_1(y - x)) + (1 - \alpha) f(x + t_2(y - x)) \\ &= \alpha g(t_1) + (1 - \alpha) g(t_2). \end{aligned}$$

In a similar fashion, we can show that if $g(t)$ is convex, then $f(x)$ is convex as well.

Thus, it follows that

$$g(1) \geq g(0) + g'(0)(1 - 0) \Rightarrow f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

Thus, we complete the proof.

(c) Show that f is convex if and only if $\nabla f(x)$ satisfies, for each x, y,

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq 0.$$

Use this property to verify if the function

$$x^T Q x, \quad Q = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

is convex.

Solution: To show convexity implies monotonicity, rearrange the terms of the previous inequality. To show monotonicity implies convexity, define $g(t) = f(x + t(y - x))$ for $t \in [0, 1]$. We have $g'(t) = \langle y - x, \nabla f(x + t(y - x)) \rangle$, and observe that $g'(1) \geq g'(0)$. If we show that $g'(t)$ is nondecreasing, then we have shown that $g(t)$ is convex.

We prove that nondecreasing functions of intervals of $\mathbb{R}$ are convex as follows: suppose $f : [a, b] \to \mathbb{R}$, and suppose $a \leq x < y \leq b$. Then, by the mean value theorem, we know that there exists a $z \in [x, y]$ such that $f'(z) = \frac{f(y) - f(x)}{y - x}$, which in turn implies $f(y) = f(x) + f'(z)(y - x)$. But we know that $f'(z) \geq f'(x)$ by the nondecreasing derivative property. Thus, we get $f(y) \geq f(x) + f'(x)(y - x)$, and hence we provce nondecreasing functions are convex.

To show $g(t)$ is convex, choose $0 \leq t_1 < t_2 \leq 1$ (they can in fact be arbitrarily close). Then,

we have

$$\begin{aligned} g'(t_2) - g'(t_1) &= \langle y - x, \nabla f(x + t_2(y - x)) - \nabla f(x + t_1(y - x))\rangle \\ &= \frac{1}{t_2 - t_1}\langle (x + t_2(y - x)) - (x + t_1(y - x)), \dots \\ &\qquad\qquad \nabla f(x + t_2(y - x)) - \nabla f(x + t_1(y - x))\rangle \\ &\geq 0 \end{aligned}$$

Thus, we see that $g(t)$ is convex. Then, we know that $g(1) \geq g(0) + g'(0)$, which is equivalent to $f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle$.

For the numerical example, observe that $Q \succ 0$. Then, $\nabla f(x) = 2Qx$, and

$$2\langle Qx - Qy, x - y\rangle = 2(x - y)^T Q(x - y) \geq 0.$$

Thus, we can conclude that $f(x) = x^T Qx$ for Q as given above is convex.

(d) Suppose $f \in C^2$. Then, show that for each x, $H_f(x) \equiv \nabla^2 f(x)$ is positive semi-definite. Furthemore, show that the converse is true as well. Use this property to check the convexity of the function given in Problem 6.

Solution: First, we show that a function with a PSD Hessian is convex. We can take the second order Taylor series and apply the mean value theorem, giving us

$$f(y) = f(x) + \langle \nabla f(x), y - x\rangle + \frac{1}{2}(y - x)^T \nabla^2 f(z)(y - x)$$

where $z \in \mathbb{R}^n$. Since $\nabla^2 f(x) \succeq 0 \; \forall \; x$, we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle.$$

The same result can also be shown using restrictions on lines. Suppose $g(t) = f(x + t(y - x))$. Then, we have $g''(t) = (y - x)^T H_f(x + t(y - x))(y - x)$. By the positive semidefinite nature of $H_f(x) = \nabla^2 f(x)$, we know that $g''(t) \geq 0$ for all t. Then, by the Taylor series expansion of $g(t)$ and the Mean Value Theorem, we get

$$g(1) = g(0) + g'(0) + \frac{1}{2}g''(t) \geq g(0) + g'(0)$$

for some $t \in [0, 1]$, thus proving the statement.

We have shown previously that a function $f : [a, b] \to \mathbb{R}$ is convex if it is nondecreasing. Here, we first show that every convex function $f : [a, b] \to \mathbb{R}$ is nondecreasing. Choose $a \leq x < y \leq b$. Then, by the convexity of f, we have

$$f(y) - f(x) \geq f'(x)(y - x) \quad \text{and} \quad f(x) - f(y) \geq f'(y)(x - y).$$

Adding the two and rearranging terms, we get

$$(f'(y) - f'(x))(y - x) \geq 0,$$

and since $y > x$, we get $f'(y) \geq f'(x)$ for all x, y. Thus, we can conclude that $f''(x) \geq 0$ for all $x \in [a, b]$. Next, define $g(t) = f(x + t(y - x))$. Since convex functions restricted to lines are convex, we have $g''(t) \geq 0$, and we know that $g''(t) = (y - x)^T H_f(x + t(y - x))(y - x)$. Thus,

it follwws that $H_f(x) \succeq 0$ for all x.

For the numerical example, recall that $f(x, y) = x^4 + y^4 - 4xy$. The Hessian is given by

$$H_f(x, y) = \begin{bmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{bmatrix}.$$

We see that, while the trace is always positive, $\det(H_f(x, y)) = 144x^2y^2 - 16 = 16(9x^2y^2 - 1)$, which can take negative values, and thus has points at which one of the eigenvalues is negative. Thus, we conclude that $f(x, y)$ is not convex.

(e) Show that the set $S := \{x : f(x) \leq t\}$, assuming S is nonempty, is convex. Verify this property for the function $f(x) = e^x$, and where $t = e$.

Solution: Suppose $x_1, x_2 \in S$. We need to show that every point on the line segment connecting x_1 and x_2 also lies in S, which means that we require

$$f(\alpha x_1 + (1 - \alpha)x_2) \leq t.$$

By the convexity of f, we have

$$f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2) \leq t.$$

12. Let S be a convex set. Show that $x^* = \arg\max_S f(x)$, where $f(x)$ is a convex function, lies on the boundary of S.

Solution: Let $z = \max_{x \in S} f(x)$, and suppose we have x, y such that $z = \alpha x + (1 - \alpha)y$. Then, by the definition of convex functions,

$$f(z) \leq \alpha f(x) + (1 - \alpha)f(y) \leq \max\{f(x), f(y)\}.$$

Thus, the only way the definition of convexity and the fact that $z = \max_{x \in S} f(x)$ are not contradicted is if z is on the boundary.